Nov 17, 2025 (Due: 08:00 Nov 24, 2025)

1. (a) Construct an example of linear system such that the Jacobi method converges while the Gauss–Seidel method diverges.
(b) Construct an example of positive definite linear system such that the Gauss–Seidel method converges while the Jacobi method diverges.

2. Let $B \in \mathbb{C}^{n\times n}$, $g \in \mathbb{C}^n$. Suppose that $\rho(B) = 0$. Show that the iterative scheme

$$x^{(k+1)} = Bx^{(k)} + g$$

converges to the solution of $x = Bx + g$ for any initial guess $x^{(0)}$ within at most n iterations.

3. Find all eigenvalues and eigenvectors of the $n \times n$ tridiagonal matrix

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{bmatrix}.$$

4. Let A, $C \in \mathbb{C}^{n\times n}$ be Hermitian and positive definite. There exists a matrix B that satisfies $A = C - B^*CB$. Show that the iterative scheme

$$x^{(k+1)} = Bx^{(k)} + g$$

converges for any initial guess $x^{(0)}$, where $g \in \mathbb{C}^n$ is a given vector.

5. Numerically solve the 2D Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

over the unit square $[0,1]^2$ with boundary conditions

$$u(0,y) = u(1,y) = u(x,1) = 0, \qquad u(x,0) = \sin(\pi x).$$

Use the Jacobi method or the Gauss–Seidel method to solve the discretized system. Visualize the solution and the convergence history.

6. (optional) The SOR method generalizes the Gauss–Seidel method by introducing a real parameter ω. The iterative scheme of the SOR method reads

$$x^{(k+1)} = (1-\omega)x^{(k)} + \omega D^{-1}\big(Lx^{(k+1)} + Ux^{(k)} + b\big).$$

Show that the SOR method diverges if $\omega \notin (0,2)$.

7. (optional) Let A be a real symmetric nonsingular matrix with positive diagonal entries. Suppose that the Gauss–Seidel method for solving $Ax = b$ converges for any initial guess. Show that A is positive definite.
(Hint: Show that $C = A - B^\top AB$ is positive definite, where $B = I - (D-L)^{-1}A$.)